

Dense Video Captioning With Early Linguistic Information Fusion

Nayyer Aafaq^{ID}, Ajmal Mian^{ID}, *Senior Member, IEEE*, Naveed Akhtar^{ID}, *Member, IEEE*, Wei Liu, *Member, IEEE*, and Mubarak Shah^{ID}, *Life Fellow, IEEE*

Abstract—Dense captioning methods generally detect events in videos first and then generate captions for the individual events. Events are localized solely based on the visual cues while ignoring the associated linguistic information and context. Whereas end-to-end learning may implicitly take guidance from language, these methods still fall short of the power of explicit modeling. In this paper, we propose a *Visual-Semantic Embedding (ViSE) Framework* that models the word(s)-context distributional properties over the entire semantic space and computes weights for all the *n*-grams such that higher weights are assigned to the more informative *n*-grams. The weights are accounted for in learning distributed representations of all the captions to construct a semantic space. To perform the contextualization of visual information and the constructed semantic space in a supervised manner, we design *Visual-Semantic Joint Modeling Network (VSJM-Net)*. The learned ViSE embeddings are then temporally encoded with a *Hierarchical Descriptor Transformer (HDT)*. For caption generation, we exploit a transformer architecture to decode the input embeddings into natural language descriptions. Experiments on the large-scale ActivityNet Captions dataset and YouCook-II dataset demonstrate the efficacy of our method.

Index Terms—Context modeling, dense video captioning, event localisation, language and vision, video captioning.

I. INTRODUCTION

DENSE video captioning is a challenging task which aims to detect all events in a video and generate grammatically correct and semantically rich descriptions for the events. A video may comprise multiple events of varying duration with possible overlaps. Consequently, the task of *dense video captioning* becomes significantly more challenging than image captioning [1]–[6] or even conventional video captioning [7], [7]–[13]. Research contributions in this exigent direction have started to emerge only recently due to the advancements in Deep Learning [14] and the availability of large-scale datasets [15]–[18].

Manuscript received 21 July 2021; revised 15 December 2021; accepted 20 January 2022. Date of publication 25 January 2022; date of current version 7 June 2023. This work is supported in part by the Australian Research Council Discovery Project under Grant DP190102443. The work of Naveed Akhtar was supported by the Office of National Intelligence National Intelligence Postdoctoral Grant NIPG-2021-001 funded by the Australian Government. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Ting Yao. (*Corresponding author: Nayyer Aafaq.*)

Nayyer Aafaq, Ajmal Mian, Naveed Akhtar, and Wei Liu are with the Department of Computer Science and Software Engineering, The University of Western Australia, Perth, WA 6009, Australia (e-mail: nayyer.aafaq@research.uwa.edu.au; ajmal.mian@uwa.edu.au; naveed.akhtar@uwa.edu.au; wei.liu@uwa.edu.au).

Mubarak Shah is with the University of Central Florida, Orlando, FL 32826 USA (e-mail: shah@crv.ucf.edu).

Digital Object Identifier 10.1109/TMM.2022.3146005

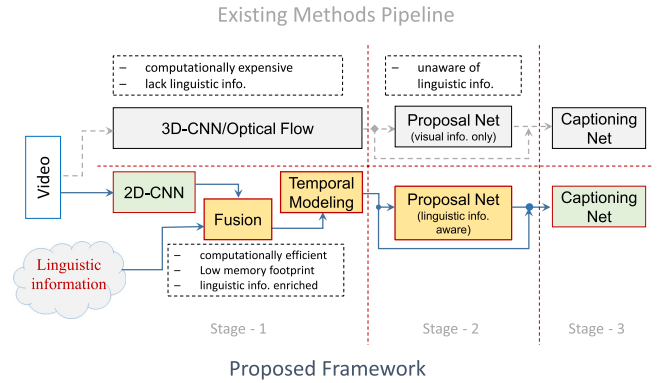


Fig. 1. *Proposed framework (bottom) vs contemporary pipeline (top).* We divide the overall framework into three stages, namely; representation learning *i.e.*, stage-1, event proposals generation *i.e.*, stage-2, and captioning net *i.e.*, stage-3. We summarize the differences of our technique at each component as compared to existing methods. For instance, in contrast to popular C3D based representation learning in stage-1, our proposed features are enriched with spatio-temporal and linguistic information before being fed to the proposals generation network. We re-design our proposals generation network to be linguistic information aware. Moreover, we presented temporal encoding technique which is faster and efficient when compared to the popular C3D-based methods.

Most contemporary video captioning approaches are confined to short videos that capture single events, which does not help the cause of dense video captioning because these methods do not address the challenges of dense captioning. For instance, *dense video captioning* must explicate the boundaries of events in a video, generate event proposals within multiple and possibly overlapping events, and describe all the detected events sequentially. Such requirements clearly distinguish this task from conventional video captioning.

Currently, the challenges of dense video captioning are addressed by dividing the task into detecting events (activities) in a video and then captioning those [17], [19]–[21]. The first sub-task is performed with event proposal modules [22]–[26], whereas the latter employs separate captioning modules. The modules performing these sub-tasks are subsequently joined to generate a dense caption [17], [19], [24]. However, there is an undesired disassociation between the two modules in this scheme. This has motivated researchers to train the framework components alternatively [17], [24] or jointly [19]–[21]. Nevertheless, these strategies induce only limited correlation between the two modules. The vast majority of existing literature focuses purely on the visual content [20], [21], [24], [27], [28], while ignoring

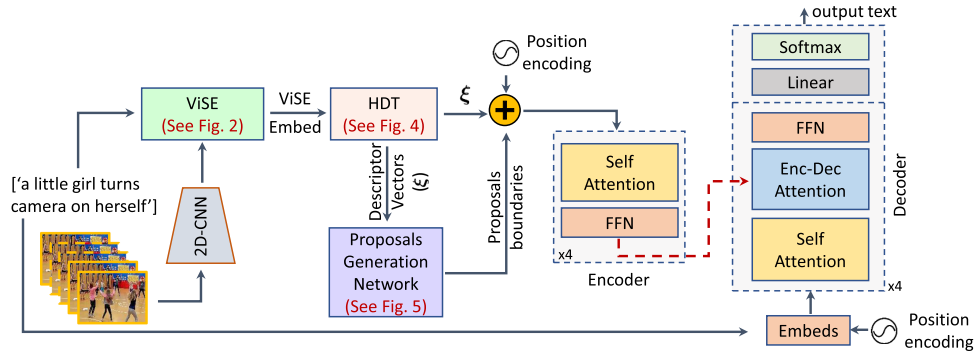


Fig. 2. *ViSE transformer DVC framework*: Visual-Semantic Embeddings (ViSE) are learned from the visual content and captions of videos and subsequently transformed with a Hierarchical Descriptor Transformer (HDT). The transformed features are used in proposal generation network. The detected proposal features are then passed to the transformer encoder and the associated word embeddings are passed to the decoder unit. The decoder further employs an encoder-decoder multi-headed attention and forwards the representations to a Fully-connected Feed-forward Network (FFN) which generates the next word with the highest probability for the caption.

the associated semantic information (see Fig. 1). However, semantic information of a video provides crucial cues for event detection and their description. For instance, in a long instructional video on ‘how to bake a cake’, events may include ‘leave dough for two hours to rise’. We only see the dough but the associated caption helps us understand that the dough has risen and is now ready to bake. This is true for all events where visual changes are subtle. Ignoring semantics, it is therefore impossible for a model to detect such events and generate the desired caption for the video.

We propose a *Visual-Semantic Embedding (ViSE)* framework that exploits visual information as well as the associated linguistic content in the event detection and caption generation process. Considering that some *n*-grams are more informative than others, we first extract *n*-grams and their associated weights from the captions. In our technique, higher weight indicates that the *n*-gram is more informative and depicts stronger bonding with the context words. These weighted *n*-grams are then processed by a *Sen2Vec* model to learn the caption embeddings and to construct the semantic space. The visual information and the semantic space are then contextualised in a supervised manner with a *Visual-Semantic Joint Embedding Network (VSJM-Net)*. The representations of *VSJM-Net* i.e., the final stage of the proposed *ViSE* framework, are temporally encoded with Short Fourier Transform in a hierarchical manner. The resulting *ViSE* representation is then passed to our proposal and captioning modules. To generate the event proposals, we propose an encoder-decoder scheme based network. In contrast to existing methods, our proposal network takes into account the associated video descriptions to localize events in a semantically meaningful manner. The proposal network generates event proposals from a video i.e., start time, end time, and confidence score of the proposal.

Dense video captioning can be treated as a machine translation problem, where input video features are translated into natural language sentences. Hence, machine translation techniques can also be leveraged for this problem. For the dense video captioning module, we employ the *Transformer* [29] architecture to generate sentences. Due to its self-attention mechanism, the transformer is able to capture long term dependencies that are present in videos for the task of dense captioning. Our overall

framework is shown in Fig. 2 and our main contributions are summarised as follows.

- 1) We model the word-contextual distributional properties and extract weighted *n*-grams from the available linguistic information. The weighted *n*-grams, with higher weights assigned to more informative *n*-grams and vice versa, are used to learn the caption embeddings which are subsequently used to construct a semantic space.
- 2) We propose a *Visual-Semantic Joint Embedding Network (VSJM-Net)* for the fusion of semantic and visual information in videos. We temporally model the contextualised information (i.e. *ViSE*) generated by the internal representation of this network with a hierarchical descriptor transformer. Such modelling results in computationally efficient and discriminative representation of the *ViSE* features.
- 3) With thorough empirical analysis, we demonstrate that semantic contextualisation in dense video captioning contributes substantially towards the caption quality. We establish the effectiveness of our technique on the large-scale ActivityNet Captions [17] dataset and YouCook-II [25] dataset.

The rest of this paper is organized as follows. Section II provides an overview of the related literature and puts our contributions into context. Section III describes the methodology in details and Section IV discusses evaluation metrics and datasets. Section V presents our results and comparisons with the existing methods, with qualitative results presented in Section VI. Section VII presents ablation studies, and the article concludes in Section VIII.

II. RELATED WORK

Below, we review the literature related to the integral components of dense video captioning followed by a brief survey of the methods designed specifically for dense video captioning.

A. Event Detection in Videos

With the recent introduction of large scale activity datasets [30], [31], research efforts has moved from simpler tasks such

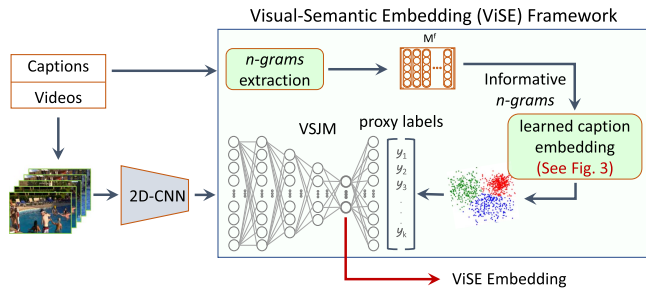


Fig. 3. *Visual-Semantic Embedding (ViSE) computation*: Semantic notions in video captions are quantitatively encoded with informatively weighted n -gram features over which a caption embedding is learned. A semantic codebook comprising proxy labels is generated with K-Means clustering applied to learned caption embeddings. The *VSJM-Net* maps features of the input video frames ('avg pool' layer activations of ResNet-152) to the proxy labels. *VSJM-Net* thus maps similar visual contents in videos to a semantic space constructed with proxy labels.

as activity classification to categorize videos *a.k.a.*, activity detection [32]–[35]. Activity detection is mainly handled as a spatio-temporal [36], [37] or temporal only detection task [23], [38]–[42]. For the former, events are localized in spatio-temporal cubes in a video. For the latter, temporal methods only focus on predicting the start and end time stamps of the activities in videos, ignoring the spatial localization of actors and objects. Predominantly, temporal methods have been extended to language in vision tasks, such as dense captioning [17] and query-based activity localization [43], [44]. Compared to the more conventional sliding window strategy for event detection [40], [45], feed-forward networks are more efficient in terms of processing videos in their entirety. The former must explicitly compute subsets of windows for event detection using techniques like dictionary learning [46] or recursive networks [23], whereas the latter [22], [23] only require a forward pass through the network. More recently, Wang *et al.* [24] employed a bidirectional recurrent network, making two passes over the video to improve the accuracy of event proposals. However, the above mentioned models only take the visual features into account while ignoring linguistic information. We believe that event detection, especially for dense video captioning, can benefit significantly from linguistic information.

B. Video Captioning

Early works in video captioning [47]–[49] employ rule-based methods where the concept is to detect subject (S), verb (V) and object (O) in a video and describe the (SVO) tuple in natural language using template based techniques. Soon template based methods became obsolete and the video captioning problem was transformed into a machine translation task [50]. With the availability of large-scale datasets [15], [16] statistical techniques were replaced by neural network based methods, with encoder-decoder architectures forming the back-bone of these techniques [11], [51]–[57]. The main idea of this line of works is to train an encoder and a decoder in an end-to-end fashion so that the encoder first encodes visual inputs with CNNs, and the decoder decodes them into natural language sentences using

RNNs. To further improve the performance of video captioning models, more recent methods augment this scheme with the advanced concepts of *e.g.* identity-aware captioning [58], attention mechanism [7], [11], semantic attribute learning [12], [13], objects and actions modeling [59]–[61], reinforcement learning [62], multimodal memory [63], [64], spatio-temporal modeling [9] and including other modalities [65], [66]. Despite superior performance, the aforementioned methods work well only for short videos, mainly comprising a single event, to describe them in a single sentence. There are very few methods that attempt to describe videos with multiple sentences. Such methods employ event proposal or captioning modules in a hierarchical manner to generate multiple sentences [28], [53], [58]. Compared to describing a short video by a single or rarely by multiple sentences, the challenges of dense video captioning are multi-fold. Hence, conventional video captioning techniques are generally ineffective for the dense captioning task.

C. Dense Video Captioning

Analogous to *dense image captioning* [67], Krishna *et al.* [17] introduced the task of *dense video captioning* that adds a temporal dimension to the problem in the form of long videos. Contemporary dense video captioning methods [17], [19]–[21], [24], [28], [68]–[73] address the challenges of this task by splitting it into two sub-tasks *i.e.* events detection and captions generation in a supervised or weakly supervised setting. The two modules include a proposal module to detect events in the input video, and a captioning module to generate the captions for the detected events. Krishna *et al.* [17] employed a multi-scale proposal generation network [23] for event detection and proposed an attention based captioning network with past and future context awareness to generate captions for all events. Wang *et al.* [24] employed a bidirectional proposal generation network to improve the detected proposals accuracy. Li *et al.* [68] proposed temporal coordinates and descriptiveness regression to localize the proposals in the video and employed an attribute-augmented captioning network [4] for improved performance. Similarly, Zhou *et al.* [19] proposed an end-to-end transformer model to address the dense video captioning problem. Ma *et al.* [74] demonstrated improvement in captioning model performance by efficiently learning higher-order interaction between the video objects. To reduce redundancy in the generated captions, Xiong *et al.* [28] proposed a method to generate captions in a progressive manner conditioned over the previous caption. More recently, Mun *et al.* [21] proposed an event sequence generation network that reduces the number of proposals and employs two-level reward maximization and generates more coherent story. Iashin *et al.* [69] explored audio and speech modalities to further improve the performance of dense video captioning framework. Fujita *et al.* [75] recently proposed an evaluation framework to cater for the story aspect of the generated dense sentences.

Attempts have also been made to tackle the problem in a weakly supervised manner to mitigate the laborious task of human annotations of videos. These methods do not require temporal segmentation for training of the dense video captioning

model. Rather, the techniques are built on an assumption that each temporal segment is described by one caption and each caption corresponds to one temporal segment. Duan *et al.* [70] adopted cyclic-consistent learning strategy and took labelled captions as input for weakly supervised dense event caption. Shen *et al.* [71] proposed multi-instance multi-label learning. This method weakly links video regions with lexical labels and generates diverse captions. Rahman *et al.* [72] incorporated audio in dense video captioning and explored its effects.

Our literature survey shows that despite the promising performances of existing methods, there exists a disassociation between event detection and description generation. Some recent methods [19], [21] try to implicitly leverage from linguistic information while joint training of the two modules, however, such an approach does not exploit the power of explicit modeling of the two modalities. Dense video captions capture details of event localization and their co-occurrence. A representation defined over both visual information in videos and semantics of their description can intrinsically couple the detection and captioning sub-tasks for dense video captioning. This novel concept forms the basis of our proposed technique, which further takes advantage of temporal modeling of visual cues for proposal representation and training of detection and captioning modules.

III. METHODOLOGY

We first discuss our *Visual-Semantic Embedding (ViSE)* framework, followed by the *Hierarchical Descriptor Transformer (HDT)* and the proposal generation network. Finally, the caption generation transformer is presented.

A. Visual-Semantic Embedding (ViSE)

Conventional dense video captioning architectures tend to comprise two loosely coupled networks *i.e.* *proposal network* and *captioning network*. In these methods, visual features of a proposal learnt from the *proposal network* are fed as input to the *decoder network* for caption generation. Hereby, we propose *Visual-Semantic Embedding (ViSE)* to jointly learn the visual and semantic space (see Fig. 3). The *ViSE* features are then fine-tuned through temporal processing before feeding to the proposal generation network and a transformer based encoder-decoder architecture.

This Visual-Semantic Embedding computation circumscribes three tasks (a) semantic space construction - where we capture word(s)-context associations for more informed *n-gram* embedding learning, (b) learning the caption and *n-gram* embedding from weighted word(s)-context, and (c) fusion of visual and learned semantics via joint modeling of the visual-semantic space.

1) *Semantic Space Construction*: According to distributional semantics, the meaning of a word can be inferred better from the aggregated context it is used in [76], [77]. Various definitions of the ‘context’ of a word and how the context ‘associates’ with the original word exist in the literature [78]. For the purpose of this research, we develop an *n-gram* based association between a word and its context by extending the traditional

TABLE I
REPRESENTATIVE EXAMPLES OF WORD-CONTEXT PAIR STRENGTH COMPUTED BY OUR METHOD FOR ACTIVITYNET CAPTIONS [17] DATASET

word-context pair	Frequency	$f(w_i, c_j)$
(‘little’, ‘girl’)	338	0.326
(‘the’, ‘girl’)	1049	0.089
(‘wella’, ‘man’)	9	0.163
(‘old’, ‘man’)	21	0.256
(‘the’, ‘man’)	80	0.054
(‘sri’, ‘lanka’)	4	0.917
(‘new’, ‘york’)	3	0.612
(‘social’, ‘media’)	3	0.760

(Top) Low strength is computed for uninformative pairs despite their high frequency of co-occurrence. Informative concepts result high association despite their low frequency.

(Bottom) Word-context pairs representing unique concepts with high association scores.

co-occurrence word association measure, Pointwise Mutual Information (PMI) [79].

Formally, let $\mathcal{W}$ define a set of unique *n-grams* including *uni-grams*, and $\mathcal{C}$ denote the set of all possible single word contexts of each element $w \in \mathcal{W}$. For clarity, we focus on *uni-grams* in the below explanation to restrict w to a single word, but the formulations are extendable to where w is a *bi-gram*, *tri-gram*, ..., or *n-gram*. Using these sets, we define a ‘word-context’ matrix $\mathbf{M}^f \in \mathbb{R}^{|\mathcal{W}| \times |\mathcal{C}|}$, where $|\cdot|$ is the cardinality of a set. The superscript f indicates that an element of our matrix is computed as $M_{[i,j]}^f = f(w_i, c_j)$ such that $f(\cdot, \cdot)$ measures the strength of association between the i^{th} word and the j^{th} context. For that matter, we exploit conditional log-likelihood of word-context pairs, and define $f(\cdot, \cdot)$ as:

$$f(w_i, c_j) = \log \frac{P(w_i, c_j)}{P(w_i)P(c_j)}, \quad (3.1)$$

where, $P(w_i, c_j)$ denotes the joint probability distribution over w_i and c_j ; and $P(w_i)$, $P(c_j)$ denote the marginal probabilities.

We can compute the joint probability over w_i and c_j as

$$P(w_i, c_j) = \frac{\psi(w_i, c_j)}{\sum_{k=1}^{|\mathcal{W}|} \sum_{l=1}^{|\mathcal{C}|} \psi(w_k, c_l)}, \quad (3.2)$$

where $\psi(w_i, c_j)$ denotes the number of co-occurrences of the i^{th} word w_i and j^{th} context c_j . The marginal probabilities of w_i and c_j can be computed as

$$P(w_i) = \frac{\sum_{j=1}^{|\mathcal{C}|} \psi(w_i, c_j)}{\sum_{k=1}^{|\mathcal{W}|} \sum_{l=1}^{|\mathcal{C}|} \psi(w_k, c_l)}, \quad (3.3)$$

$$P(c_j) = \frac{\sum_{i=1}^{|\mathcal{W}|} \psi(w_i, c_j)}{\sum_{k=1}^{|\mathcal{W}|} \sum_{l=1}^{|\mathcal{C}|} \psi(w_k, c_l)}. \quad (3.4)$$

To ensure that no negative values are recorded for the word-context pairs that never co-occurred, we let $f(w_i, c_j) = \max\{f(w_i, c_j), 0\}$. To demonstrate the effectiveness of $f(\cdot, \cdot)$, we show representative examples of the association strengths computed by our scheme in Table I. As can be noticed, despite the ‘high’ frequency of uninformative co-occurrences, our method results in low values of $f(w_i, c_j)$, which is exactly the

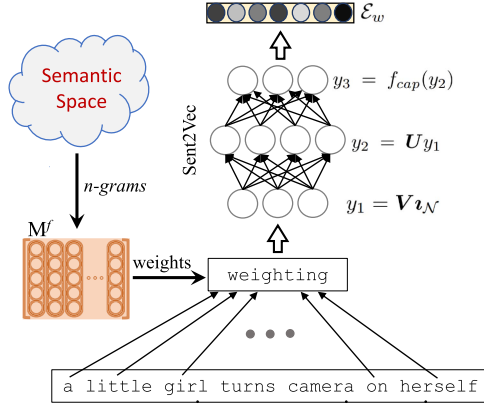


Fig. 4. n -grams and their corresponding weights are used to learn the sentence/caption embeddings. The weights correspond to the word-context association where higher values are assigned to more informative n -grams irrespective of their occurrence frequency. The whole sentence is used as a context while predicting the next n -gram.

opposite for the ‘low’ frequency informative co-occurrences. Interestingly, this method also computes high strengths for word-context pairs that represent unique concepts, e.g. social media. The definition provided in (3.1) is for a single word, i.e. $n = 1$. For the general case of n -grams, we capture the relationship as

$$f((w_1, w_2, \dots, w_n), c_j) = \log \frac{P((w_1, w_2, \dots, w_n), c_j)}{P(w_1, w_2, \dots, w_n)P(c_j)}. \quad (3.5)$$

2) *Caption Embedding Computation*: A sentence can be a combination of *uni*-grams and n -grams, which requires combining their embeddings for its representation. Simple arithmetic combination (e.g. summation, mean) of Word Embeddings (e.g. BERT [80] and its variations or XLNET [81]) is currently the most popular technique for obtaining vector representation of n -grams or sentences in the captioning literature [82]. However, to ensure the meaningfulness of captions in the semantic space, we desire that the learnt embeddings to be more selective towards content-rich n -grams and captions. Therefore, we devise a meaningfulness measure based on the co-occurrence word-context associations (i.e. M^f), which is subsequently used in our technique to increase the embedding weights of more informative n -grams.

To compute the embedding of a caption while explicitly learning the embeddings of n -grams (opposed to averaging or summing up word’s embeddings), we adopt a cultivated method inspired by FastText [83] (Fig. 4). Analogous to the use of a set of character n -grams to represent the words in FastText, we employ a set of word n -grams to represent the captions. To that end, we learn the caption embedding with the help of the Sent2Vec model [84]. In our approach, the resulting embeddings are not just learned with *uni*-grams, but also with n -grams. The Sent2Vec model solves an optimization problem that takes the form

$$\min_{U, V} \sum_{cap \in \mathcal{C}} f_{cap}(UV \mathbf{u}_N), \quad (3.6)$$

where $U \in \mathbb{R}^{k \times h}$ and $V \in \mathbb{R}^{h \times |\mathcal{V}|}$ are parameter matrices, $|\mathcal{V}|$ denotes the vocabulary size and h is the dimension of the learned

vectors (we let $k = |\mathcal{V}|$), $f_{cap}(\cdot)$ is the cost function and $\mathcal{N}$ denotes the list of n -grams, $\mathbf{u}_N \in \{0, 1\}^{|\mathcal{V}|}$ is a binary indicator vector to encode $\mathcal{N}$. The constituent words of a sentence and their combinations (i.e. n -grams) are specifically optimized over the input caption. The embedding $\mathcal{E}_c$ of a caption is finally computed as:

$$\mathcal{E}_c = \frac{1}{|\mathcal{N}|} \sum_{w \in \mathcal{N}} \mathcal{E}_w, \quad (3.7)$$

where $\mathcal{E}_w$ is the n -gram embedding that includes *uni*-grams.

3) *Visual-Semantic Space Joint Modeling*: To jointly model the visual and semantic space, our proposed VSJM-Net learns a representation which is concurrently defined over the visual contents and the semantic space constructed by the videos and their captions, respectively. The proposed network maps visual contents to a quantitative representation of semantic notions. To that end, the learned weighted caption/sentence embeddings (i.e. $\mathcal{E}_c$ in (3.7)) are first clustered using K-means to generate proxy labels for the supervised training of the VSJM-Net. Next, the centroids of the resulting clusters are used to determine the labels in the semantic space of the captions. The proposed network learns a non-linear mapping of the visual contents onto these labels. As depicted in Fig. 3, the VSJM-Net is a deep fully connected neural network that consists of Q -layers, where $Q = 5$ in our experiments. For a given training sample $\mathcal{R}(x_j^i) \in \mathbb{R}^{2000}$, where $\mathcal{R}(x_j^i)$ is the feature vector of j^{th} frame in the i^{th} cluster i.e., x_j^i , the output of the first layer is $\mathbf{h}^{(1)} = f(\mathbf{W}^{(1)}\mathcal{R}(x_j^i) + \mathbf{b}^{(1)}) \in \mathbb{R}^{p^{(1)}}$, where $\mathbf{W}^{(1)} \in \mathbb{R}^{p^{(1)} \times 2000}$ is a weight matrix to be learned in the first layer, $\mathbf{b}^{(1)} \in \mathbb{R}^{p^{(1)}}$ is a bias vector and $f(\cdot)$ represents non-linear activation function. Similarly, the output of the q^{th} layer $\mathbf{h}^{(q)}$ is computed as;

$$\mathbf{h}^{(q)} = f(\mathbf{W}^{(q)}\mathbf{h}^{(q-1)} + \mathbf{b}^{(q)}) \in \mathbb{R}^{p^{(q)}},$$

where $q = 2, 3, \dots, Q$. Lastly, the output of the last fully-connected layer $\mathbf{h}^{(Q)}$ is passed through a softmax layer to find the appropriate label. For simplicity, we represent the softmax output of the VSJM-Net as $g(x_j^i)$. In line with the provided details, the loss function is computed as;

$$\mathcal{L}(\Theta, \mathcal{X}^c) = \mathbb{E}_{\mathcal{X}^c} \left[\frac{1}{KL_i} \sum_{i=1}^K \sum_{j=1}^{L_i} \ell(y_j^i, g(x_j^i)) \right], \quad (3.8)$$

where ℓ denotes softmax loss function (that computes softmax cross entropy between output of softmax i.e., $g(x_j^i)$ and proxy labels) and y_j^i is the proxy label of i^{th} cluster i.e., $y_j^i = i$. The VSJM-Net learns semantically contextualized visual representation of the visual contents to extract features from video frames. We use the layer $p^{(4)}$ to produce a series of $m = 256$ neuron activations per video frame for further processing.

B. Hierarchical Descriptor Transformer

We extract the visual-semantic features, say $\mathcal{F}$ (i.e. ViSE Embeddings) for all frames processed by ViSE. These features are further processed by a *Hierarchical Descriptor Transformer (HDT)*. The HDT employs hierarchical Fourier transformation [9], [85] to capture temporal dynamics of frame-level

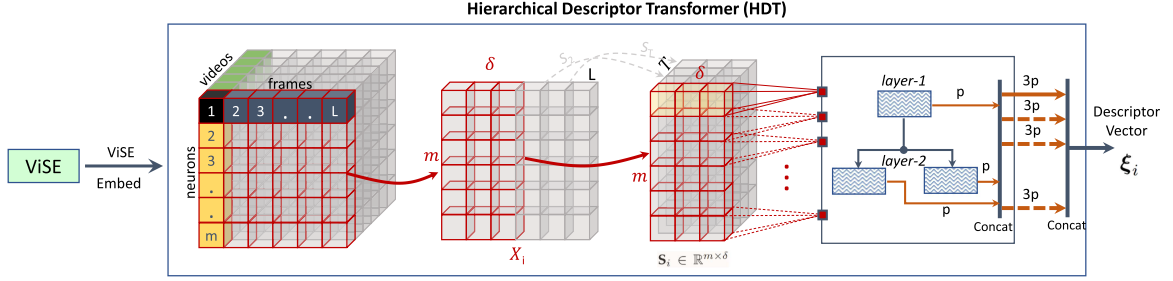


Fig. 5. *Hierarchical Descriptor Transformer (HDT)*: Learned visual-semantic representation from ViSE is temporally modelled by the HDT. Neuron-wise frame features are processed with hierarchical Fourier transformations, and ‘ p ’ coefficients from each layer in the hierarchy are captured (*i.e.* $3p$ coefficients per neuron per frame stream or ‘ δ ’ frames). The process is repeated for m neurons. The ‘ $3p$ ’ coefficients for ‘ m ’ neurons are concatenated to compute the final descriptor vector ξ_i .

features (see Fig. 5). We recursively partition the video-level features obtained from ViSE into halves and process each segment with Fourier transform to capture the temporal dynamics of all the events in a video. The final descriptive representation is a concatenation of Fourier coefficients from all segments. The proposed *HDT* captures the temporal dynamics of our time series data without training.

For a more concrete understanding, let $\mathcal{F} = \{f_1, f_2, \dots, f_L\}$ denote the L ViSE features of a frame sequence for a video X_i , where $f_i \in \mathbb{R}^m$. We use a temporal resolution of $\delta = 64$ and create a set of visual streams $\mathcal{S} = \{S_1, S_2, \dots, S_T\}$ using the feature set, where $T = |\mathcal{F}|/\delta$. Here, $S_i \in \mathbb{R}^{m \times \delta}$ is a matrix, with each column representing the activations of *VSJM-Net*. Each stream of $\mathcal{S}$ *i.e.*, S_i can be treated as a function of time. To each stream, we hierarchically apply the Fourier transformation at n levels to obtain its coefficients. We extract only ‘ $3p$ ’ low frequency coefficients per neuron for ‘ m ’ neurons to form the desired features. It was empirically discovered that the number of low frequency coefficients gauge the extent of cancellation for similar visual content in the consecutive frames. The final feature for each stream is the concatenation of the ‘ $3p$ ’ low frequency coefficients for all the neurons. For ‘ T ’ streams in a video X_i , the computed descriptor vectors becomes $\xi_i = \{\xi_1, \dots, \xi_T\}$. These semantically enriched spatio-temporal features are subsequently processed by our proposal generation and captioning modules.

C. Proposals Generation Network

The main role of a proposal generation network in a dense video captioning pipeline is to detect temporal boundaries of a video within which events may have occurred. In contrast to existing methods, our proposal module explicitly accounts for the visual as well as semantic information to detect the proposal boundaries. Moreover, our method exploits both past and future contexts in a video. First, we encode the frame-level features in the enriched representations that integrate semantic clues under the ViSE framework. Second, we design a *Video Level Sequence Encoder (VLSE)* that processes this information across the frames to detect event boundaries in a video (Fig. 6).

For a given video, the VLSE receives a series of temporally refined ViSE features *i.e.* $\xi_i = \{\xi_1, \dots, \xi_T\}$. Using this stream, it detects events in the video in both backward and forward directions to

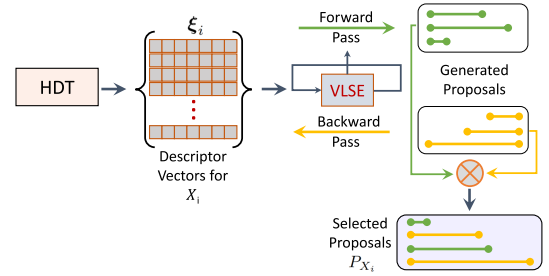


Fig. 6. For a given video, visual-semantic representation of all visual streams from the HDT is processed in forward and backward directions for proposal detection. The proposals collected from each direction are fused using a multiplicative strategy. Proposals with confidence score greater than a threshold are finally selected for the captioning network training.

account for the past and future. Inspired by [24], we fuse the confidence scores resulting from both directions to get the final confidence scores of the detected events. For the i^{th} video X_i , we obtain a set of event proposals P_{X_i} for description generation, such that $P_{X_i} = \{p_{X_i}^j | p_{X_i}^j = (start_{X_i}^j, end_{X_i}^j, score_{X_i}^j), \forall j\}$.

We differ in two ways from the existing techniques that employ proposals generation network in dense captioning. Firstly, our encoder is based on the ViSE framework followed by the HDT transformation, as discussed in the previous sections. The common practice is to directly use visual features from C3D [86] for the encoder. Our encoder is much more sophisticated and results in better performance (see § IV). Secondly, we explicitly incorporate the semantic information in our encoder which is generally ignored by others while encoding the proposal network input content. To the best of our knowledge, this is a new concept in the context of dense video captioning.

D. Captions Generation Transformer

The role of mapping the ViSE features of a given proposal to captions is fulfilled by a Transformer in our technique. Originating in the machine translation literature [29], transformers are known for their efficient and effective encoding and decoding of long sequences. Our transformer encoder takes the sequence of visual stream descriptors from the HDT discussed above, *i.e.* $\xi_i = \{\xi_1, \xi_2, \dots, \xi_T\}$ temporally corresponding to proposals P_{X_i} from proposals generation network. Using this representation and word embeddings from the caption, our decoder

generates an output representation g , which is subsequently used with the vocabulary to predict the probability of the next word in the caption, *i.e.* $P(w_{t+1}|g)$. Before discussing the encoder and decoder of our technique, we first describe the transformer backbone concepts *i.e.* *scaled dot-product attention* and *multi-head attention*. The input to *scaled dot-product attention* operation consists of queries $q \in \mathbb{R}^{d_k}$, keys $k \in \mathbb{R}^{d_k}$, and values $v \in \mathbb{R}^{d_v}$. The process computes dot product of each query q_i with all the keys k_t where $t = 1, 2, \dots, \tilde{T}$ and outputs the weighted sum of the values by applying a softmax function. We also apply a scaling factor of $\frac{1}{\sqrt{d_k}}$ to contain the dot product magnitude and subsequently avoid small gradients. Let $K \in \mathbb{R}^{d_k \times \tilde{T}}$ and $V \in \mathbb{R}^{d_v \times \tilde{T}}$ be the matrices resulting from arranging the keys and values, respectively. The output for the *scaled dot-product attention* is

$$A_{tt}(q_i, K, V) = \left(\frac{\exp \{K^\top q_i / \sqrt{d_k}\}}{\sum_{t=1}^{\tilde{T}} \exp \{k_t q_i / \sqrt{d_k}\}} \right) V. \quad (4.1)$$

The *multi-headed attention* can be defined as an extension of the scaled dot-product attention, where it allows the model to jointly attend information in parallel. We preserve the computational complexity by reducing the dimensions by a factor of number of heads, *i.e.* $d = d_k/h$, where h is number of parallel layers or “heads”. Each head can be defined as

$$head_h(q_i, K, V) = A_{tt}(q_i W_h^{q_i}, K W_h^K, V W_h^V), \quad (4.2)$$

where $W_h^{q_i}, W_h^K, W_h^V \in \mathbb{R}^{\frac{d_k}{h} \times d_k}$ represent independent head trainable matrices, where $h = 1, 2, \dots, H$. The *multi-head attention* is a concatenation of all the independent heads with trainable parameter matrix $W^o \in \mathbb{R}^{d_k \cdot H \times d_k}$. The output for the multi-head attention module can thus be expressed as:

$$\hat{A}_{tt}(q_i, K, V) = \begin{pmatrix} head_1(q_i, K, V) \\ head_2(q_i, K, V) \\ \dots \\ head_H(q_i, K, V) \end{pmatrix} W^o \quad (4.3)$$

Besides *multi-headed attention*, we also use a 2-layer fully connected network with ReLU activations for all the layers of encoder and decoder blocks. We denote this network as a function $\Psi(\cdot)$ in the text to follow, where we describe the encoder and decoder units of our technique.

1) *Encoder*: The encoder comprises L layers, and it receives a set of descriptor vectors that temporally corresponds to it respective proposal. Let us denote this set associated with the j^{th} proposal as $\{\xi_j\}$. We normalize the input of each layer of the encoder following [87]. That is,

$$\bar{\eta}_j^l = \Phi(\mathcal{T}_e(\{\xi_j^l\})), \quad (4.4)$$

where $\mathcal{T}_e$ represents the internal representation of the encoder layer and Φ represents the layer normalisation. Each encoder layer comprises two sub-layers *i.e.* *multi-headed attention* and the *2-layer fully connected network* discussed above. The representation learned by the l^{th} encoder layer in our network can be expressed as:

$$\bar{\rho}_j^l = \Phi(\Psi(\Phi(\hat{A}_{tt}(\bar{\eta}_j^l, \bar{\eta}_j^l, \bar{\eta}_j^l) + \eta_j^l))). \quad (4.5)$$

$$\eta_j^{l+1} = \varphi_j^l + \hat{A}_{tt}(\bar{\eta}_j^l, \bar{\eta}_j^l, \bar{\eta}_j^l) + \eta_j^l. \quad (4.6)$$

Notice the self attention in (4.5) where the multi-headed attention uses identical queries, keys and values. Each time-step in the current encoder layer accounts for context from all states from the previous layer simultaneously. This gives an advantage over the traditional RNN which uses context from the last state only.

2) *Decoder*: The decoder stacks similar layer structure as the encoder. However, it additionally performs *multi-headed attention* over the encoder output. Input to the decoder comprises of output from the encoder η_j^l along with the set of word embeddings w^j at the time step t . The input is fed forward to L decoder layers, where each subsequent layer takes input from the previous layer. Similar to the encoder, our decoder also incorporates layer normalisation and residual connections. The decoder outputs a representation ζ_j^l that is further utilized in the next word generation.

Formally, the states for l^{th} decoder layer can be represented as follows:

$$\bar{\zeta}_j^l = \Phi(\mathcal{T}_d(w_j^l; \forall w \leq t)), \quad (4.7)$$

$$\bar{\rho}_j^l = \Phi(\hat{A}_{tt}(\bar{\zeta}_j^l, \bar{\zeta}_j^l, \bar{\zeta}_j^l) + \zeta_j^l), \quad (4.8)$$

$$\Upsilon_j^l = w_j^l + \hat{A}_{tt}(\bar{\zeta}_j^l, \bar{\eta}_j^l, \bar{\eta}_j^l), \quad (4.9)$$

$$\bar{\zeta}_j^{l+1} = \Upsilon_j^l + \Psi(\Upsilon_j^l); \forall \leq t, \quad (4.10)$$

where $\mathcal{T}_d$ represents the internal representation of the decoder layer and η_j^l is the encoder output. Equation (4.8) refers to *self multi-headed attention* whereas (4.9) depicts the *encoder-decoder multi-headed attention* as it attends both encoder output and decoder's previous states. At time t , given the decoder representation ζ_j^l , the generator computes a probability distribution over the words $p(w_{t+1}|\zeta_j^l)$. The word with the highest probability in the distribution is selected as the next word.

IV. EVALUATION

A. Datasets

We evaluate our technique using the large-scale ActivityNet Captions dataset [17] and YouCook-II dataset [25] for dense video captioning.

1) *ActivityNet Captions*: dataset [17] comprises $\sim 100k$ natural language sentences describing $\sim 20k$ untrimmed long videos of real world scenes. These videos are a subset of the larger ActivityNet dataset [30]. Each video consists of at least two, and average three annotated events along with start and end time for each event. The videos span over almost 849 hours in total. On-average, an annotated sentence in the dataset contains 13.48 words. Each video in the dataset is annotated by multiple sentences, covering almost 94.6% of the visual content, whereas every description covers ~ 36 seconds of the video content. Furthermore, there is almost 10% temporal overlap of the events in the video that makes the dataset particularly interesting and challenging. The dataset is split into 50%, 25%, and 25% portions comprising 10024/4926/5044 videos for training, validation and testing, respectively. Ground truth annotations of the

TABLE II

EVENT DETECTION PERFORMANCE OF THE PROPOSAL NETWORK WITH *ViSE* EMPLOYED AS THE *ENCODER*. RESULTS ARE AVERAGED ON FOUR THRESHOLDS $\{0.3, 0.5, 0.7, 0.9\}$ FOR TEMPORAL INTERSECTION OVER UNION (tIoU) FOR THE ACTIVITYNET CAPTIONS VALIDATION SET. f AND b DENOTES 'FORWARD' AND 'BACKWARD' PASS OF THE VIDEO RESPECTIVELY

Method	Proposal Network		Avg. Recall	Avg. Prec.	F1-Score
	(Visual Encdr.)	(Seq. Decdr.)	(@1000)	(@1000)	(@1000)
MFT [28]	Two-stream CNNs	LSTM	0.243	0.514	0.331
SST [22]	C3D	GRU (f)	0.910	0.411	0.566
Bi-SST [24]	C3D	LSTM ($f + b$)	0.875	0.459	0.602
SDVC [21] [‡]	C3D	GRU + ESGN	0.556	0.576	0.566
<i>ours</i>	w/o <i>ViSE</i>	LSTM ($f + b$)	0.625	0.414	0.498
	with <i>ViSE</i>	LSTM ($f + b$)	0.883	0.474	0.617

[‡]Denotes very low average number of proposals (*i.e.* 2.85) as compared to other listed methods.

test split are withheld for competitions and online server evaluation.

2) *YouCook-II*: dataset [25] comprises ~ 2000 open domain videos covering 89 cooking recipes. The total time span of all the videos is ~ 175.6 hours with associated semantics that encapsulate a vocabulary of 2600 words. Each video in the dataset is divided into 3-16 segments, elaborating cooking steps. All the segments are semantically annotated and temporally time-stamped for localisation. The average length of each video is 316 seconds, reaching up to the maximum duration of ~ 600 seconds. The length of each video segment varies from 1 s to 264 seconds. Following [25], we split the dataset into training, validation and testing sets with the ratio 66%:23%:10%.

B. Implementation Details:

For supervised training of the *VSJM-Net*, we first extract the frame-wise features of videos from the activations of '*avgpool*' layer of *ResNet-152* [88]. From the available semantic information, we first extract *uni-grams*, *bi-grams*, *tri-grams*, and *4-grams* along with their weights. The weights correspond to contextual bonding and informativeness of the corresponding *n-gram*. These weighted *n-grams* are used to learn the sentence embeddings with 700 dimensions each. The embeddings are grouped into K clusters to get the same number of labels for *VSJM-Net* training. We select $K = 2000$ and 200 for ActivityNet Captions and YouCook-II datasets respectively. Selection of ' K ' is a trial-and-error process. We approximate this number using the *DBSCAN* algorithm that is able to capture data association and structure by varying its *minPoints* and *epsilon* parameters such that minimal outliers are produced. We use *minPoints* = 3 and $\epsilon = 0.85$ in our approximation process. We then pass the ' K ' to *K-means* algorithm to generate clusters. The performance of *VSJM-Net* is subjective and is not highly sensitive to ' K '. During *VSJM-Net* training, dropout is set to 0.5 for better generalisation while the network is trained for 20 epochs with a batch size of 250. From the trained *VSJM-Net*, activations of $p^{(4)}$ are used to generate the time series signals for the subsequent temporal refinement. Since the number of Fourier coefficients directly control the intensity of cancellation of similar visual content, we empirically select the first $p = 2$

low frequency coefficients in each of the two-level hierarchical transformation. The output feature is then reduced to 1024 dimensions with PCA. For video level sequence encoder (VLSE), we employ LSTM to sequentially encode the descriptor vectors in forward as well as backward direction. We set the threshold for selected proposals to be 0.8 for captioning.

For our Transformer model, we use the model dimension $d = 1024$ and the hidden size of feed-forward network layers as 2048. As too many heads also deteriorate the quality of the model, we set the number of heads $H = 4$. We use the *Adam* [89] optimizer with dynamic learning rate, initially 10^{-3} , reduced by a factor of 10 after the first 10 epochs. We set $\beta_1 = 0.9$, $\beta_2 = 0.98$, and smoothing parameter $\gamma = 0.7$. For model regularisation, we use dropout [90] at the output of each sub-layer before its addition and normalisation. Dropout is also used in the additions of embeddings and positional encodings, both in encoder and decoder stacks. We generate maximum 1000 proposals per video and pass the features of the high confidence proposals ($\text{tIoU} \geq 0.8$) for caption generation. Later, we organise the generated captions and associated proposal pairs based on their scores in descending order to select the top 100 proposal/caption pairs for evaluation. We employ TensorFlow framework for the development with a NVIDIA Titan Xp GPU.

V. RESULTS AND COMPARISONS

A. Temporal Events Proposal Localisation

We first establish the effectiveness of our framework for the proposal detection task. In this case, our proposal module uses the *ViSE* framework as the encoder, but adopts the decoder of *Bi-SST* [24]. Given that the anchor size largely impacts the model performance, for a fair comparison, we maintain the same anchor sizes and other experimental settings for all the models in comparison.

1) *Compared Techniques*: We compare the performance of our event detection module with state-of-the-art methods and present quantitative results in Table II. In the table, *SST* forward passes a video once and generates up to 1000 proposals. *Bi-SST* builds upon *SST* and processes the video twice, in forward and backward manner, to cater for both the past and future contexts. It then, fuses scores of both passes to detect the event proposals. In Table II, we are taking advantage over *Bi-SST* by using its decoder but our *ViSE* encoder, followed by *HDT*.

2) *Evaluation Metrics*: We use the commonly employed metrics; average Recall R@1000, Precision P@1000 averaged at tIoU thresholds $\{0.3, 0.5, 0.7, 0.9\}$. Additionally, we also compute the F1 score that simultaneously accounts for precision and recall and is proven to be a more reasonable metric for event detection accuracy.

2) *Performance*: In Table II, *SST* shows the highest recall score, however, its precision is the lowest. As dense video captioning accounts for all the generated proposals, low precision proposals get penalised severely. This deteriorates the model performance. Hence, a better indicator of a generated proposal quality is the F1 score that accounts for both precision and recall. We outperform all other methods on this metric. Our method outperforms *SST* in recall and F1 score by 13.3% and 8.3%

TABLE III
DENSE VIDEO CAPTIONING RESULTS ON [17] WITH LEARNED PROPOSALS ON THE UPDATED METRIC

Model	B1	B2	B3	B4	M	C	R	SODA _(C)
Krishna's [17]	10.81	4.57	1.90	0.71	5.69	12.43	-	-
TDA-CG [24]	10.75	5.06	2.55	1.31	5.86	7.99	-	3.15
Masked Transformer [19]*	9.96	4.81	2.42	1.15	4.98	9.25	-	4.02
SDVC [w/o RL] [21] [†]	13.76	6.36	3.0	1.41	6.92	25.49	-	-
MDVC [69]*	-	-	2.60	1.07	7.31	-	-	-
Rahman's [72]	10.0	4.20	1.85	0.90	4.93	13.79	10.39	-
ViSE + Transformer	14.44	7.05	3.60	1.77	7.56	14.98	13.68	5.16

*Indicates the use of additional modalities.

[†]Indicates score on very few (average 2.85) proposals.

‘-’ Indicates scores not reported.

respectively. Despite the fact that *SDVC* generates very few proposals that contribute towards high recall and precision, we outperform *SDVC* in F1 score by a significant margin of 9.1%. Further, we also outperform *Bi-SST* by achieving an overall 1% gain in recall, 3.16% gain in precision, and 2.4% improvement in the F1 score. We also highlight the performance gap between two variants of our model, *i.e.*, with *ViSE* and *without ViSE*. The former incorporates linguistic information for the event detection, whereas the latter does not. We can see clearly that incorporating the linguistic information at early stage boosts the model performance.

B. Dense Captioning Evaluation

We compare the overall proposed scheme of *ViSE Transformer* with several existing state-of-the-art techniques, including Krishna's [17], Duan's [70], JeDDI-Net [20], DVC [68], Masked Transformer [19], TDA-CG [24], *SDVC* [21], Rahman's [72] and MDVC [69].

2) *Evaluation Metrics*: We use four widely used metrics; BLEU@N [91], METEOR [92], CIDEr [93], and ROUGE_L [94]. We refer to [18] for the definitions and comparisons of these metrics. The evaluation metrics for dense captioning consider the proposal as well as captioning accuracy. If a proposed event does not overlap its ground truth location with more than a predefined *temporal Intersection over Union (tIoU)*, it is scored as *zero*. We use the official evaluation script¹ provided by [17] to evaluate the detected proposals and generated captions score. To evaluate the proposals accuracy, we measure the recall and precision score of the generated proposals. Here too, we report the metric scores as averages taken over the tIoU thresholds of 0.3, 0.5, 0.7, and 0.9 with the ground truth proposals. For dense video description, the evaluation metrics have also been updated recently. The earlier (old) version does not consider penalty for incorrect proposals and hence results in overestimated performance. For a clear benchmarking, we report results on both old and updated versions of the evaluation metrics. We further evaluate models with a newly proposed evaluation metric, Story Oriented Dense video cAptioning evaluation framework (SODA) [75].

2) *Performance Comparison*: In Table III, we report our results using the updated evaluation metrics where, instead of one

out of multiple incorrect predictions for a video, the remaining incorrect ones are also account for. Hence, the scores are generally lower for all methods. The updated metrics scores are being gradually adopted in recent papers. A clear correlation between the updated and old metric scores can be observed. Our technique outperforms most of the techniques on multiple metrics. Note that, *SDVC* [21] originally reports results after optimising METEOR and CIDEr scores using additional reinforcement learning (RL) on the generated captions. We report *SDVC* results before RL application for a fair comparison with other methods. Moreover, [21] generates very few proposals *i.e.* ~ 2.85 , compared to an average ~ 89 proposals per video by our method. The evaluation scores of all metrics in general and CIDEr in particular, are inversely proportional to the number of proposals being evaluated per video. This still gives an additional advantage to *SDVC*. Additionally, we also report score on SODA_(C) to capture the story aspect of the predicted captions. As SODA score is sensitive to the number of proposal-caption pairs, for a fair comparison, we report the baseline results for the total number of captions equal to the number of ground-truth event captions, *i.e.*, $m = 1$. The results are shown in the last column of the Table III. As evident from the table, our proposed model consistently outperforms the competitors.

In Table IV, we report results of different variants of our technique for a clear analysis of the strengths of the various modules of the proposed approach. The ‘Base’ model (*i.e.*, Base (C3D)-LSTM (U)) is implemented using 3D-CNN to extract input video features. This variant does not use contextualization and temporal refinement, and the events are directly detected using 3D-CNN features. The LSTM (U) indicates a Uni-directional LSTM at the decoder stage. The ‘ResNet (MP)-LSTM (U)’ employs the strategy of Mean Pooling (MP) the 2D-ResNet features to aggregate temporal information in the frames, and perform event detection. It is apparent from the results that a 2D-CNN representation followed by mean pooling is more desirable than the 3D-CNN representation alone. Hence, we eventually selected the activations of 2D-ResNet's *avgpool* layer to encode the input videos. The ‘ResNet (TE)-LSTM (U)’ variant replaces the mean pooling strategy with the Temporal Encoding (TE) proposed in this work using the short Fourier Transform in the HDT. Finally, the ‘ViSE-LSTM’ and ‘ViSE-Transformer’ variants use both the temporal encoding and the visual semantic contextualization. For the former, LSTM (B) indicates a Bi-directional LSTM. Using ViSE with transformer, *i.e.* ‘ViSE-Transformer’

¹[Online]. Available: https://github.com/ranjaykrishna/densevid_eval

TABLE IV
DENSE VIDEO CAPTIONING RESULTS OF VARIOUS VARIANTS OF OUR METHOD ON ACTIVITYNET CAPTIONS [17]

Method	B1	B2	B3	B4	M	C	R
Base (C3D)-LSTM (U)	14.19	3.86	1.11	0.38	7.07	5.12	14.84
ResNet (MP)-LSTM (U)	17.54	5.20	1.94	0.82	8.73	7.25	17.94
ResNet (TE)-LSTM (U)	18.72	7.80	3.61	1.87	9.05	12.41	19.58
ViSE-LSTM (U)	19.89	8.98	4.35	2.09	9.45	12.48	19.97
ViSE-LSTM (B)	20.59	9.11	4.58	2.14	9.59	15.58	20.33
ViSE + Transformer	22.18	10.92	5.58	2.72	10.78	23.89	21.98

Average scores of BLEU (B1-B4), METEOR (M), CIDEr (C), ROUGE (R) are reported across tIoU of 0.3, 0.5, 0.7, 0.9. MP, TE, U, and B denote *Mean Pool* strategy, *Temporal Encoding*, *Uni-Directional*, and *Bi-Directional* respectively.

TABLE V
DENSE VIDEO CAPTIONING RESULTS ON ACTIVITYNET CAPTIONS [17] WITH GROUND TRUTH PROPOSALS

Method	B3	B4	M	C	R
Krishna's [17]	7.12	3.98	9.46	24.56	-
TDA-CG [24]	-	-	10.89	-	-
Masked Transformer [19]*	5.69	2.67	11.06	-	-
DVC [68]*	4.55	1.62	10.33	25.24	-
Rahman's [72]*	3.04	1.46	7.23	25.36	15.37
ViSE + Transformer	7.86	3.97	12.89	26.52	25.37

Average scores of BLEU (B1-B4), METEOR (M), CIDEr (C), ROUGE (R) are reported across tIoU of 0.3, 0.5, 0.7, 0.9.

* Indicates use of additional modalities (e.g. optical flow, audio).

‘-’ Indicates scores are not reported by the method.

TABLE VI
DENSE VIDEO CAPTIONING RESULTS ON YOU-COOK-II DATASET [25] WITH THE LEARNED PROPOSALS

Method	B4	M	C	R	SODA _(C)
MT [19]	0.33	2.44	6.84	5.91	-
ViSE + Transformer	1.09	4.31	9.07	10.51	4.21

Average scores of BLEU B4, METEOR (M), CIDEr (C), ROUGE (R) are reported across tIoU of 0.3, 0.5, 0.7, 0.9.

outperforms all other variants of our technique. Note that, in our framework, three components *i.e.*, 2D-CNN, ViSE and HDT work hand-in-hand to process a video, jointly providing an alternate for video processing with 3D-CNN. Hence, it is not meaningful to simply replace 2D-CNN with 3D-CNN in our setup because the framework is designed to separately account for the temporal dimension with HDT.

Table V reports results of our technique using the ground truth proposals. This specifically highlights performance of the captioning module as ground truth proposals are fed to the model. As visible in Table V, we achieve the best performance for three metrics while maintaining highly competitive performance across the remaining metrics.

In Table VI, we report the performance of our model on the YouCook-II dataset. No dense captioning results have been reported to date on this dataset using the updated evaluation metrics. Although the dataset is less challenging in terms of temporal events detection as the segments follow the procedural recipe, it is more challenging due to small objects that are hard to detect using abstract visual features but are important to be in the captions. For comparison, we re-evaluated [19] on the updated metrics and to be fair with our method, we also take the top 89

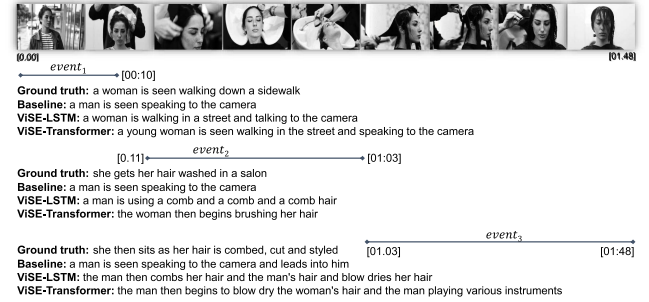


Fig. 7. Qualitative results of dense video captioning from ActivityNet captions [17] dataset. The generated sentences of final model are compared with Baseline and ViSE-LSTM variants of the model. The bars indicate the events with starting and ending time mentioned against each. The example video shows a woman getting her hair styled in a beauty parlor.

proposals from [19] to compute the metric scores. As shown in the table, our technique outperforms [19] with significant margins on all metrics. Additionally, we also report the SODA_(C) score of our method in the table. For consistency, the score for YouCook-II dataset is reported with $m = 1$, *i.e.*, number of captions selected are equal to number of ground truth captions per video.

VI. QUALITATIVE RESULTS AND ANALYSIS

We provide representative qualitative results from ActivityNet Captions and YouCook-II datasets in Figs. 7 and 8, respectively. We compare the captions generated by various variants of our technique to illustrate the role of different modules namely Baseline (visual-only), ViSE-LSTM (ViSE features with Bi-directional LSTM), and ViSE-Transformer (ViSE features with Transformer).

VII. DISCUSSION

We report the details on the number of parameters and training time with different compositions of our framework in Table VII where PropNet, Trans. and P(M) respectively abbreviate the ‘proposal network,’ ‘transformer’ and ‘parameters in Millions’. A check-mark indicates that the module is a part of the overall framework. The train time is indicated as minutes per epoch for training on a NVIDIA Titan Xp GPU. The reported values indicate that the ViSE-LSTM still has a smaller memory footprint compared to simply using the popular C3D network for input video representation followed by the proposal network

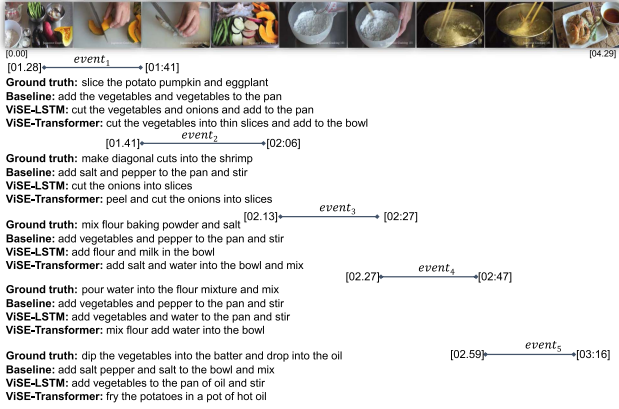


Fig. 8. Qualitative results of dense video captioning from YouCook-II [25] dataset. The generated sentences of final model are compared with Baseline and ViSE-LSTM variants of the model. The bars indicate the events with starting and ending time mentioned against each. The example video shows a Japanese recipe of cooking (involves subtle actions such as cutting, peeling and frying) vegetables and shrimp.

TABLE VII

MEMORY FOOTPRINT, P(M) PARAMETERS IN MILLIONS OF OUR METHOD. CHECK-MARK INDICATES THE MODEL INCLUDES THE RESPECTIVE NETWORK AND TRAINING TIME IS IN MIN. PER EPOCH

Model	C3D	ResNet	PropNet	TE	ViSE	LSTM	Trans.	P(M)	Time
Krishna's [17]	✓	-	✓	-	-	✓	-	91.5	~ 240
SDVC [21]	✓	-	✓	-	-	✓	-	176.0	~ 444
Base(c)	✓	-	✓	-	-	✓	-	86.2	~ 50
ResNet(MP)	-	✓	✓	-	-	✓	-	66.4	~ 50
ResNet(TE)	-	✓	✓	✓	-	✓	-	66.4	~ 60
ViSE+LSTM	-	✓	✓	✓	✓	✓	-	78.9	~ 75
ViSE+Transformer	-	✓	✓	✓	✓	-	✓	129.0	~ 71

and the captioning network. The transformer variant has the most parameters, however, its training time is still comparable to the other LSTM based variants. The proposed framework has a smaller memory footprint and computational cost as compared to the state-of-the-art methods. For instance, SDVC [21] is computationally ~6 times more expensive and has a larger memory footprint than our method. In Table VII, we calculate the number of parameters for [17] based on the model information, as the actual number of model parameters is not publicly available.

During the experiments we observed that evaluation metrics are influenced by the number of proposals generated for the captions. The effect of number of proposals on the evaluation metrics, for the same overall set of the generated captions, is inversely proportioned. Hence, increasing the number of proposals adversely affects the scores of all four metrics. The effect surfaced after evaluation code on the server was fixed and metrics were updated. In old evaluation metrics, only one out of multiple incorrect predictions for each video was counted leading to performance overestimation of earlier methods and the number of proposals generated were ineffective. However, this is not the case for the updated metrics where incorrect proposals are penalised towards the final performance of the model. Techniques that compute very few proposal, such as SDVC [21] (avg. ~2.85 proposals) are able to boost their metric scores based on limiting the number of generated proposals.

VIII. CONCLUSION

We proposed a method that strengthens the coupling between event detection and caption generation for dense video captioning by incorporating semantic information in event detection and generates captions with semantically enriched features. Our technique first models the word(s)-context distributional properties over the entire semantic space and extracts weights for all the n -grams. With the weighted n -grams, we learn distributed representations of all captions to construct the semantic space. We then perform contextualization of the visual information and the constructed semantic space in a self-supervised manner. For that matter, we proposed a *Visual-Semantic Embedding (ViSE)* framework. The *ViSE* embedding are transformed into descriptor vectors with a *Hierarchical Descriptor Transformer (HDT)*. For caption generation, we exploit transformer architecture to decode the *ViSE* representations into natural language descriptions and also compare it with its LSTM-based counterpart. Experiments on the large-scale ActivityNet Captions and YouCook-II datasets establish the effectiveness of our method.

ACKNOWLEDGMENT

The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the Army Research Office or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for Government purposes not with standing any copyright notation herein.

REFERENCES

- [1] M. Yang et al., "Multitask learning for cross-domain image captioning," *IEEE Trans. Multimedia*, vol. 21, no. 4, pp. 1047–1061, Apr. 2018.
- [2] Y. Feng, L. Ma, W. Liu, and J. Luo, "Unsupervised image captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 4120–4129.
- [3] X. Yang, K. Tang, H. Zhang, and J. Cai, "Auto-encoding scene graphs for image captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 10677–10686.
- [4] T. Yao, Y. Pan, Y. Li, Z. Qiu, and T. Mei, "Boosting image captioning with attributes," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 4894–4902.
- [5] L. Gao, Z. Guo, H. Zhang, X. Xu, and H. T. Shen, "Video captioning with attention-based LSTM and semantic consistency," *IEEE Trans. Multimedia*, vol. 19, no. 9, pp. 2045–2055, Sep. 2017.
- [6] Y. Wang, Z. Lin, X. Shen, S. Cohen, and G. W. Cottrell, "Skeleton key: Image captioning by skeleton-attribute decomposition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 7272–7281.
- [7] C. Yan et al., "STAT: Spatial-temporal attention mechanism for video captioning," *IEEE Trans. Multimedia*, vol. 22, no. 1, pp. 229–241, Jan. 2019.
- [8] J. Zhang and Y. Peng, "Video captioning with object-aware spatio-temporal correlation and aggregation," *IEEE Trans. Image Process.*, vol. 29, pp. 6209–6222, 2020.
- [9] N. Afaq, N. Akhtar, W. Liu, S. Z. Gilani, and A. Mian, "Spatio-temporal dynamics and semantic attribute enriched visual encoding for video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 12487–12496.
- [10] S. Liu, Z. Ren, and J. Yuan, "SibNet: Sibling convolutional encoder for video captioning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 9, pp. 3259–3272, Sep. 2021.
- [11] L. Yao et al., "Describing videos by exploiting temporal structure," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 4507–4515.
- [12] Z. Gan et al., "Semantic compositional networks for visual captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 5630–5639.

- [13] Y. Pan, T. Yao, H. Li, and T. Mei, "Video captioning with transferred semantic attributes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 984–992.
- [14] Y. LeCun, Y. Bengio, and G. Hinton, "Deep Learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [15] S. Gella, M. Lewis, and M. Rohrbach, "A dataset for telling the stories of social media videos," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2018, pp. 968–974.
- [16] L. Zhou, Y. Kalantidis, X. Chen, J. J. Corso, and M. Rohrbach, "Grounded video description," in *Proc. Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 6578–6587.
- [17] R. Krishna, K. Hata, F. Ren, L. Fei-Fei, and J. C. Niebles, "Dense-captioning events in videos," in *Proc. Int. Conf. Comput. Vis.*, 2017, pp. 706–715.
- [18] N. Afaq, A. Mian, W. Liu, S. Z. Gilani, and M. Shah, "Video description: A survey of methods, datasets, and evaluation metrics," *ACM Comput. Surv.*, vol. 52, no. 6, pp. 1–37, 2019.
- [19] L. Zhou, Y. Zhou, J. J. Corso, R. Socher, and C. Xiong, "End-to-end dense video captioning with masked transformer," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 8739–8748.
- [20] H. Xu, B. Li, V. Ramanishka, L. Sigal, and K. Saenko, "Joint event detection and description in continuous video streams," in *Proc. IEEE Conf. Appl. Comput. Vis.*, 2019, pp. 396–405.
- [21] J. Mun, L. Yang, Z. Ren, N. Xu, and B. Han, "Streamlined dense video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 6588–6597.
- [22] S. Buch, V. Escorcia, C. Shen, B. Ghanem, and J. C. Niebles, "SST: Single-stream temporal action proposals," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 2911–2920.
- [23] V. Escorcia, F. C. Heilbron, J. C. Niebles, and B. Ghanem, "DAPs: Deep action proposals for action understanding," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 768–784.
- [24] J. Wang, W. Jiang, L. Ma, W. Liu, and Y. Xu, "Bidirectional attentive fusion with context gating for dense video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7190–7198.
- [25] L. Zhou, C. Xu, and J. J. Corso, "Towards automatic learning of procedures from web instructional videos," in *Proc. Conf. Artif. Intell.*, 2018, pp. 7590–7598.
- [26] Y. Zhao *et al.*, "Temporal action detection with structured segment networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2914–2923.
- [27] C.-Y. Ma *et al.*, "Attend and interact: Higher-order object interactions for video understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6790–6800.
- [28] Y. Xiong, B. Dai, and D. Lin, "Move forward and tell: A progressive generator of video descriptions," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 468–483.
- [29] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [30] F. Caba Heilbron, V. Escorcia, B. Ghanem, and J. Carlos Niebles, "ActivityNet: A large-scale video benchmark for human activity understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 961–970.
- [31] A. Gorbun *et al.*, "THUMOS challenge: Action recognition with a large number of classes," 2015. [Online]. Available: <http://www.thumos.info/>
- [32] A. Gaidon, Z. Harchaoui, and C. Schmid, "Temporal Localization of Actions With Actoms," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2782–2795, 2013.
- [33] R. Hou, C. Chen, and M. Shah, "Tube convolutional neural network (t-cnn) for action detection in videos," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 5822–5831.
- [34] G. Gkioxari and J. Malik, "Finding action tubes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 759–768.
- [35] Y. Tian, R. Sukthankar, and M. Shah, "Spatiotemporal deformable part models for action detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2013, pp. 2642–2649.
- [36] G. Yu and J. Yuan, "Fast action proposals for human action detection and search," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 1302–1311.
- [37] P. Weinzaepfel, Z. Harchaoui, and C. Schmid, "Learning to track for spatio-temporal action localization," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 3164–3172.
- [38] B. Singh, T. K. Marks, M. Jones, O. Tuzel, and M. Shao, "A multi-stream bi-directional recurrent neural network for fine-grained action detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 1961–1970.
- [39] S. Yeung, O. Russakovsky, G. Mori, and L. Fei-Fei, "End-to-end learning of action detection from frame glimpses in videos," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 2678–2687.
- [40] Z. Shou, D. Wang, and S.-F. Chang, "Temporal action localization in untrimmed videos via multi-stage CNNs," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 1049–1058.
- [41] A. Montes, A. Salvador, S. Pascual, and X. Giro-i Nieto, "Temporal activity detection in untrimmed videos with recurrent neural networks," in *Proc. 1st NIPS Workshop Large Scale Comput. Vis. Syst.*, 2016, pp. 1–5.
- [42] S. Ma, L. Sigal, and S. Sclaroff, "Learning activity progression in lstms for activity detection and early detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 1942–1950.
- [43] J. Gao, C. Sun, Z. Yang, and R. Nevatia, "TALL: Temporal activity localization via language query," in *Proc. Int. Conf. Comput. Vis.*, 2017, pp. 5267–5275.
- [44] L. A. Hendricks *et al.*, "Localizing moments in video with natural language," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 5803–5812.
- [45] J. Gao, Z. Yang, K. Chen, C. Sun, and R. Nevatia, "TURN TAP: Temporal unit regression network for temporal action proposals," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 3628–3636.
- [46] F. C. Heilbron, J. C. Niebles, and B. Ghanem, "Fast temporal activity proposals for efficient detection of human actions in untrimmed videos," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 1914–1923.
- [47] A. Kojima, T. Tamura, and K. Fukunaga, "Natural language description of human activities from video images based on concept hierarchy of actions," *Int. J. Comput. Vis.*, vol. 50, no. 2, pp. 171–184, 2002.
- [48] P. Das, C. Xu, R. F. Doell, and J. J. Corso, "A thousand frames in just a few words: Lingual description of videos through latent topics and sparse object stitching," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2013, pp. 2634–2641.
- [49] N. Krishnamoorthy, G. Malkarnenkar, R. J. Mooney, K. Saenko, and S. Guadarrama, "Generating natural-language video descriptions using text-mined knowledge," in *Proc. Conf. Artif. Intell.*, vol. 1, 2013, p. 2.
- [50] M. Rohrbach *et al.*, "Translating video content to natural language descriptions," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2013, pp. 433–440.
- [51] S. Venugopalan *et al.*, "Sequence to sequence-video to text," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 4534–4542.
- [52] S. Venugopalan *et al.*, "Translating videos to natural language using deep recurrent neural networks," in *Proc. NAACL: Hum. Lang. Technol.*, 2015, pp. 1494–1504. [Online]. Available: <https://www.aclweb.org/anthology/N15-1173>
- [53] H. Yu, J. Wang, Z. Huang, Y. Yang, and W. Xu, "Video paragraph captioning using hierarchical recurrent neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 4584–4593.
- [54] L. Baraldi, C. Grana, and R. Cucchiara, "Hierarchical boundary-aware neural encoder for video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 1657–1666.
- [55] C. Hori *et al.*, "Attention-based multimodal fusion for video description," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 4193–4202.
- [56] J. Donahue *et al.*, "Long-term recurrent convolutional networks for visual recognition and description," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 2625–2634.
- [57] X. Wu, G. Li, Q. Cao, Q. Ji, and L. Lin, "Interpretable video captioning via trajectory structured localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 6829–6837.
- [58] J. S. Park, T. Darrell, and A. Rohrbach, "Identity-aware multi-sentence video description," in *Proc. Eur. Conf. Comput. Vis.*, 2020.
- [59] B. Pan *et al.*, "Spatio-temporal graph for video captioning with knowledge distillation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 10867–10876.
- [60] Q. Zheng, C. Wang, and D. Tao, "Syntax-aware action targeting for video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 13093–13102.
- [61] Z. Zhang *et al.*, "Object relational graph with teacher-recommended learning for video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 13275–13285.
- [62] X. Wang, W. Chen, J. Wu, Y.-F. Wang, and W. Y. Wang, "Video captioning via hierarchical reinforcement learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 4213–4222.
- [63] J. Wang, W. Wang, Y. Huang, L. Wang, and T. Tan, "M3: Multimodal memory modelling for video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7512–7520.
- [64] W. Pei *et al.*, "Memory-attended recurrent network for video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 8347–8356.

- [65] W. Hao, Z. Zhang, and H. Guan, "Integrating both visual and audio cues for enhanced video caption," in *Proc. Conf. Artif. Intell.*, 2018, pp. 6894–6901.
- [66] J. Xu, T. Yao, Y. Zhang, and T. Mei, "Learning multimodal attention lstm networks for video captioning," in *Proc. 25th ACM Int. Conf. Multimedia*, 2017, pp. 537–545.
- [67] J. Johnson, A. Karpathy, and L. Fei-Fei, "DenseCap: Fully convolutional localization networks for dense captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 4565–4574.
- [68] Y. Li, T. Yao, Y. Pan, H. Chao, and T. Mei, "Jointly localizing and describing events for dense video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7492–7500.
- [69] V. Iashin and E. Rahtu, "Multi-modal dense video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, 2020, pp. 958–959.
- [70] X. Duan *et al.*, "Weakly supervised dense event captioning in videos," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 3059–3069.
- [71] Z. Shen *et al.*, "Weakly supervised dense video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 5159–5167.
- [72] T. Rahman, B. Xu, and L. Sigal, "Watch, listen and tell: Multi-modal weakly supervised dense event captioning," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2019, pp. 8908–8917.
- [73] S. Chen, W. Jiang, W. Liu, and Y.-G. Jiang, "Learning modality interaction for temporal sentence localization and event captioning in videos," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 333–351.
- [74] C.-Y. Ma *et al.*, "Attend and interact: Higher-order object interactions for video understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 6790–6800.
- [75] S. Fujita, T. Hirao, H. Kamigaito, M. Okumura, and M. Nagata, "SODA: Story oriented dense video captioning evaluation framework," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 517–531.
- [76] Z. S. Harris, "Distributional structure," *Word*, vol. 10, no. 2-3, pp. 146–162, 1954.
- [77] J. R. Firth, "A synopsis of linguistic theory, 1930-1955," *Stud. linguistic Anal.*, 1957.
- [78] P. D. Turney and P. Pantel, "From frequency to meaning: Vector space models of semantics," *J. Artif. Intell. Res.*, vol. 37, pp. 141–188, 2010.
- [79] K. W. Church and P. Hanks, "Word association norms, mutual information, and lexicography," in *Proc. Anterior Cruciate Ligament*, 1989, pp. 76–83.
- [80] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. 16th Annu. Conf. North Amer. Chapter Assoc. Comput. Linguistics*, 2019, pp. 4171–4186.
- [81] Z. Yang *et al.*, "XLNet: Generalized autoregressive pretraining for language understanding," in *Proc. Neural Inf. Process. Syst.*, 2019, pp. 5753–5763.
- [82] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating text generation with bert," in *Proc. Int. Conf. Learn. Representations*, 2020.
- [83] P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov, "Enriching word vectors with subword information," in *Proc. Trans. Assoc. Comput. Linguistics*, 2017, pp. 135–146.
- [84] M. Pagliardini, P. Gupta, and M. Jaggi, "Unsupervised learning of sentence embeddings using compositional n-Gram features," in *Proc. 16th Annu. Conf. North Amer. Chapter Assoc. Comput. Linguistics*, 2018, pp. 528–540.
- [85] A. V. Oppenheim, *Discrete-Time Signal Processing*. Santiago, Chile: Pearson, 1999.
- [86] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3D convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 4489–4497.
- [87] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *Statistics*, vol. 1050, no. 21, 2016.
- [88] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [89] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Representations*, 2015.
- [90] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.
- [91] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. Meeting Assoc. Comput. Linguistics*, 2002, pp. 162–166.
- [92] S. Banerjee and A. Lavie, "METEOR: An automatic metric for MT evaluation with improved correlation with human judgments," in *Proc. Anterior Cruciate Ligament Workshop*, 2005, pp. 65–72.
- [93] R. Vedantam, C. L. Zitnick, and D. Parikh, "Cider: Consensus-based image description evaluation," in *IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 4566–4575.
- [94] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Proc. Text Summarization Branches Out*, 2004, pp. 74–81.



Nayyer Aafaq received the B.E. degree (Hons.) in avionics from the College of Aeronautical Engineering, National University of Sciences and Technology (NUST), Islamabad, Pakistan, in 2007, and the M.S. degree (Hons.) in systems engineering from the Queensland University of Technology, Brisbane, QLD, Australia, in 2012. He is currently working toward the Ph.D. degree with the School of Computer Science and Software Engineering, The University of Western Australia, Perth, WA, USA. From 2007 to 2011, he was a Research Assistant with STG Research Institute, Islamabad, Pakistan, and from 2013 to 2017, a Lecturer with NUST. His research in computer vision and pattern recognition has been published in prestigious venues of the field including the IEEE CVPR and ACM Computing Surveys. His current research interests include deep Learning, video analysis and intersection of natural language processing, computer vision, and machine learning. He was the recipient of the SIRF Scholarship at UWA.



Ajmal Mian (Senior Member, IEEE) received the Ph.D. degree (with distinction) from the The University of Western Australia, Perth, WA, Australia, in 2006. He is currently a Professor of computer science with The University of Western Australia. He has authored or coauthored more than 200 scientific papers in reputable journals and conferences. His research interests include computer vision, machine learning, adversarial deep learning, 3D shape analysis, video understanding, and multimodal biometrics. He was the recipient of the Australasian Distinguished Doctoral Dissertation Award from Computing Research and Education Association of Australasia, prestigious Australian Postdoctoral and Australian Research Fellowships in 2008 and 2011 respectively, UWA Outstanding Young Investigator Award 2011, West Australian Early Career Scientist of the Year 2012 award, Vice-Chancellors Mid-Career Research Award 2014, Aspire Professional Development Award 2016 and Excellence in Research Supervision Award 2017. He has secured eight Australian Research Council grants, one National Health and Medical Research Council Grant, two U.S. Dept. of Defense grants and a DAAD German Australian Research Cooperation Grant. He is an Associate Editor for the IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *Pattern Recognition Journal*.



Naveed Akhtar (Member, IEEE) received the Ph.D. degree in computer science from The University of Western Australia (UWA), Perth, WA, Australia, and the master's degree from Hochschule Bonn-Rhein-Sieg, Sankt Augustin, Germany. He is currently an Assistant Professor with UWA. He has authored or coauthored in the prestigious sources of computer vision, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE and IEEE Conference on Computer Vision and Pattern Recognition. His research interests include adversarial machine learning, explainable artificial intelligence, multiple object tracking, action recognition, and hyperspectral image analysis. He was the recipient of the prestigious fellowship by the Australian Office of National Intelligence, a finalist of the Western Australia's Early Career Scientist of the Year 2021, and Universal Scientific Education and Research Network top ten young scientist in Formal Sciences for 2021. He is an ACM Distinguished Speaker and an Associate Editor for the IEEE ACCESS and a Guest Editor of the *Neural Computing and Applications* and *Remote Sensing* journals. He is also an Area Chair of CVPR'22.



Wei Liu (Member, IEEE) received the Ph.D. degree from the University of Newcastle, Callaghan, NSW, Australia in 2003. She is currently with the Department of Computer Science and Software Engineering, The University of Western Australia, Perth, WA, Australia, and Co-lead the faculty's Big Data Research Group. She has authored or coauthored several journals and conferences, including *ACM Computer Surveys*, *Journal of Data Mining and Knowledge Discovery*, *Knowledge and Information Systems*, International Conference on Data Engineering, and ACM

International Conference on Information and Knowledge Management. Her research impact in the field of knowledge discovery from natural language text data is evident by a series of highly cited papers, and the reputable top data mining and knowledge management journals and conferences. Her current research interests include deep learning methods for knowledge graph construction from natural language text, sequential data mining, and text mining. She has won three Australian Research Council Grants and several industry grants.



Mubarak Shah (Life Fellow, IEEE) is currently the Trustee Chair Professor of computer science, founding Director with the Center for Research in Computer Vision, University of Central Florida, Orlando, FL, USA. His research interests include video surveillance, visual tracking, human activity recognition, visual analysis of crowded scenes, video registration, and UAV video analysis. He is the Editor of the international book series on video computing, Editor-in-Chief of the *Journal of Machine Vision and Applications*, and an Associate Editor for the *ACM*

Computing Surveys Journal. He was the Program Co-chair of the CVPR 2008, an Associate Editor for the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, and a Guest Editor of the special issue of the *International Journal of Computer Vision on Video Computing*. He is an ACM Distinguished Speaker. He was an IEEE Distinguished Visitor Speaker during 1997–2000. He was the recipient of the IEEE Outstanding Engineering Educator Award in 1997, Harris Corporations Engineering Achievement Award in 1999, TOKTEN awards from UNDP in 1995, 1997, and 2000, Teaching Incentive Program Award in 1995 and 2003, Research Incentive Award in 2003 and 2009, Millionaires Club Awards in 2005 and 2006, University Distinguished Researcher Award in 2007, Honorable mention for the ICCV 2005 Where Am I? Challenge Problem, and was nominated for Best Paper Award at the ACM Multimedia Conference in 2005. In 2006, he was awarded a Pegasus Professor Award, the highest award at UCF. He is a fellow of the AAAS, IAPR, and SPIE.